

Employment

Graduate Research Assistant	NC State University	Apr 2025 – Present
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- Collaborated with plant biologists and geneticists to deploy cloud-native genomic data platform on GCP (Cloud Run) with JBrowse 2 genome browser, enabling 100+ researchers to visualize 500GB+ genome datasets.
- Improved ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.

Product Engineer Intern	Isomer AI	May 2025 – May 2026
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- Shipped per-tenant AI workflow platform for insurance carriers: Temporal.io durable orchestration, LLM risk signal detection across Claude/GPT-4o/Gemini, Auth0 RBAC, MCP servers; 30+ deployments.
- Built MCP servers (Python + TypeScript); Playwright + Tailscale VPN for customer-internal browser automation.

Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
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- Partnered with healthcare data teams to build 5+ AI workflows including health claim automation using OCR, NLP, and ML pipelines; improved document processing accuracy by 25% across 10K+ daily medical claims.
- Automated AWS and GCP CI/CD pipelines; integrated Grafana monitoring and led model validation and red-teaming sessions.

AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
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- Led cross-functional SDLC for data-driven enterprise ticket analysis using NLP classification models; worked with support teams to reduce resolution time by 15% and eliminate 2,000+ monthly manual triages.
- Built Kafka + Spark + Hadoop pipelines with data engineers; cut ticket-processing latency by 20% at peak load.

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
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- Master of Science in Computer Science
- Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering, Programming Languages, Computer Networks, Distributed Systems.

Mumbai, India	Mumbai University	Aug 2020 – May 2024
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- Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infrastructure, DS & Algorithms, Machine Learning, Blockchain, Computer Networks, Database Systems.

Projects

Inventory Reservation Service	Java 21 + Spring Boot REST API	2026
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- Built a Java 21 + Spring Boot inventory reservation API: PostgreSQL row-level locking (SELECT FOR UPDATE) across multi-SKU reservations, idempotent POST endpoints, automated expiry via scheduled cleanup job.
- Testcontainers integration tests verified concurrency: qty-1 race produces exactly one success, one 409 Conflict.

KubeServe	Kubernetes LLM Inference Platform	2025
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- Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving; circuit breaker + EWMA load balancing.
- Deployed via Helm + ArgoCD GitOps on Kubernetes; load tested k6 to 500 VUs; 88 pytest integration tests.

AlgoPulse	Distributed Competitive Coding Platform	2025
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- Built AlgoPulse, a real-time competitive coding platform: Temporal.io workflows, Judge0 sandboxed code execution, Kafka event streaming, Redis pub/sub live leaderboard; 2,000 req/sec throughput.
- Deployed on AWS with auto-scaling; k6 load tested to 10,000 concurrent users; sub-100ms leaderboard latency.

Languages and Technologies

- **Languages:** Java, Python, TypeScript, JavaScript, SQL, Ruby, C
- **AI/ML:** LLMs (Claude, GPT-4o, Gemini), MCP tooling, Temporal.io, RAG, embeddings, PyTorch, scikit-learn
- **Frameworks:** Spring Boot, FastAPI, Node.js, NestJS, Next.js, React, React Native, Express
- **Cloud & Infra:** AWS, GCP, PostgreSQL, Redis, Kafka, Docker, Kubernetes, Terraform, GitHub Actions

Additional Experience and Awards

- **Open Source:** Contributed to Expertiza Rails platform (5,000+ users, 300+ contributors); added 110+ RSpec tests, increased coverage by 25%, reduced redundancy by 35%.
- **Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons; shipped 3 full-stack MVPs end-to-end within 48-hour competitive sprints.
- **Publications:** Co-authored CV-Based Cybersecurity Threat Detection (Springer 2023) and Mapify (IEEE 2022); presented to 500+ global attendees.
- **GDSC Lead:** Led Google Developer Student Club; organized 4+ tech events (hackathons, conferences) reaching 500+ students.